PRODUCT USER MANUAL

For Arctic Ocean Tidal Analysis and Forecasting Product

ARCTIC_ANALYSISFORECAST_PHY_TIDE_002_015

Issue: 1.0

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RECORD TABLE

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1. INTRODUCTION

a) Summary

ARCTIC_ANALYSISFORECAST_PHY_TIDE_002_015 is the nominal product of the Arctic Monitoring and Forecasting Center for ocean physics which contains tides, storm surge and wave signals. The product is composed of two-dimensional (2D), quarter-hourly instantaneous fields of sea surface height, and x- and y-velocity components at the sea surface.

b) Polar Projections

All products are interpolated onto the same polar stereographic output grid, although at different resolutions. HYCOM uses a staggered Arakawa C grid but all variables are defined at the center of the output grid cell after interpolation.
The same projection is used for all Arctic MFC grids on a spherical Earth, following this definition under the proj4 projection library: (Beware, this is not the WGS84 ellipsoid reference).

projection_stereo:proj4 = "+units=m +proj=stere +a=6378273.0 +b=6378273.0 +lon_0=-45.0 +lat_0=90.0 +lat_ts=90.0 +ellps=sphere" ;

This polar stereographic projection also defines the reference direction of vectors (current and sea ice velocity), which should therefore be rotated if the data is reprojected to a Lat-Long or any other grid.
In ArcGIS and QGIS softwares, the x- and y- coordinates units should be converted from 100 km to meters.

A simple code (in Matlab) is provided below :

```
nx=size(lon,1);
ny=size(lon,2);
radian=pi/180.;
dlon_ip1=(lon(2:nx,:)-lon(1:nx-1,:)).*cos( radian*.5*(lat(2:nx,:)+lat(1:nx-1,:)) );
dlat_ip1=lat(2:nx,:)-lat(1:nx-1,:);
dlon_ip1(nx,:)=dlon_ip1(nx-1,:);
dlat_ip1(nx,:)=dlat_ip1(nx-1,:);
theta_ip1=atan2(dlat_ip1,dlon_ip1); % Angle displacement with constant latitude
lines
theta_jp1=theta_ip1+90*radian;
%%rotated Easterly and Northerly components
tmpu=u.*cos(theta_ip1) + v.*cos(theta_jp1);
tmpv=u.*sin(theta_ip1) + v.*sin(theta_jp1);

A cleaner Python example

import numpy as np
import cartopy.crs as ccrs
import cartopy.feature as cfeature
from cartopy.crs import Projection
from netCDF4 import Dataset

# load data (grid and ocean velocities)
with Dataset("20191214_qhr-metno-MODEL-topaz6-tide-ARC-b20191205-fv02.0.nc")
as ds:
      vx = ds["vxo"][0]
```

```
        vy = ds["vyo"][0]
        x = ds["x"][:]
        y = ds["y"][:]
        lon = ds["longitude"][:]
        lat = ds["latitude"][:]
spd = np.hypot(vx, vy)
x2d, y2d = np.meshgrid(x, y)

# Define projections
proj_stere = ccrs.Stereographic(central_latitude=90, true_scale_latitude=90,
                                central_longitude=-45,
globe=ccrs.Globe(semimajor_axis=6378273, semiminor_axis=6378273))
proj_geo = ccrs.PlateCarree()  # lon-lat

# transform velocities from stereographic projection to lon-lat
usi, vsi  = proj_geo.transform_vectors(proj_stere, x2d, y2d, vx, vy)

# transform velocities back from lon-lat to stereographic projection
vx_, vy_  = proj_stere.transform_vectors(proj_geo, lon, lat, u, v)
```

c) Vertical coordinates

All ARCTIC ocean products – except the sea ice product - are computed on a hybrid vertical coordinate. This is a combination of isopycnal (constant density levels) in the open, stratified ocean, terrain-following coordinates in shallow coastal areas and z-levels in the mixed layer or unstratified deep ocean. See figure below (http://www.hycom.org/hycom/overview).

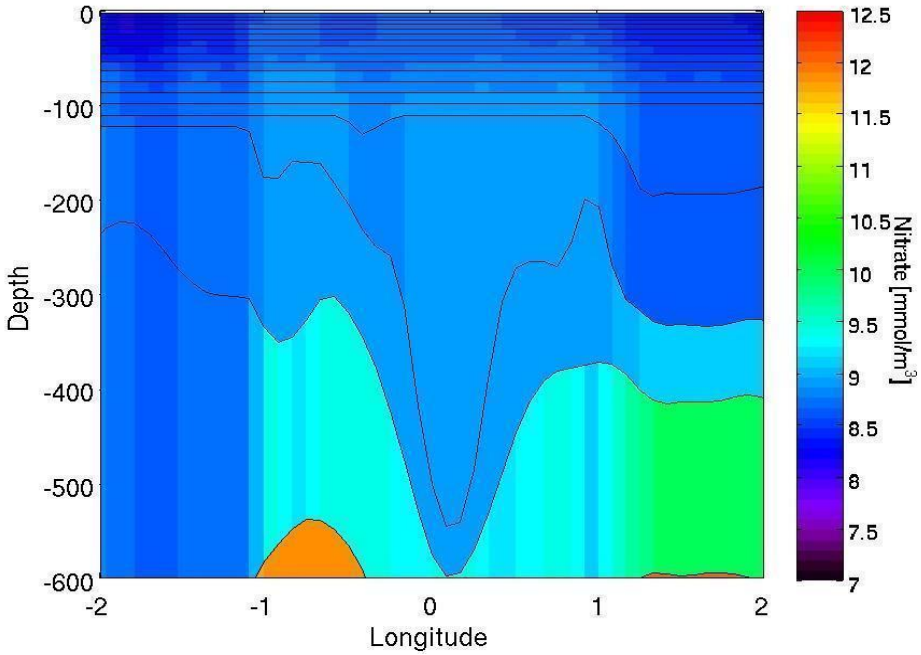


Figure 6 Vertical section of nitrates and vertical levels. Shading indicates concentrations, thin lines indicate layer interfaces. The coordinates revert from isopycnal to z-levels in the mixed layer.

Although the model performs full 3D calculation, the PHY TIDE product provides only surface variables described in 2) c) dataset details.

2. DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

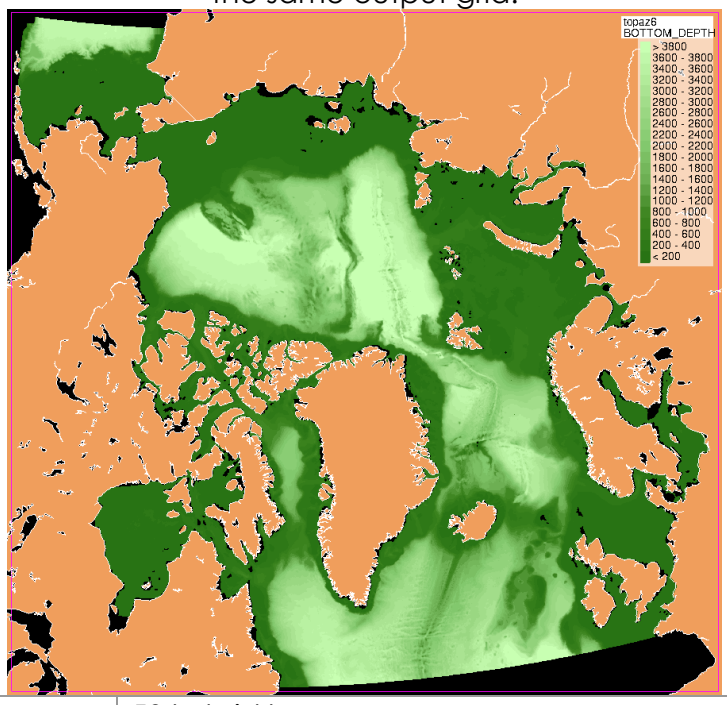
Product Lines	ARCTIC_ANALYSISFORECAST_PHY_TIDE_002_015
Geographical coverage	North of 63°N and ice-covered North Atlantic Ocean
Variables	• Sea Surface Height (includes tidal elevation) • Horizontal velocity at the sea surface (also includes tidal signals and Stokes Coriolis Drift*)
Product Type	Near Real Time - Analysis and Forecast
Available time series	-201/01/2018 to 10 Days- forecast
Temporal resolution	quarter-hourly instantaneous surface fields
Update Frequency	Daily
Target delivery time	daily10-day forecast: daily at 01:00 UTC
Horizontal resolution	3 km
Number of vertical levels	only surface variables
Format	NetCDF4

**: The surface currents are formally "total surface currents" although the standard name is kept as "ocean currents" to avoid disrupting the aggregation server.*

b) Production System Description

ARCTIC (180°W-180°E; 63N – 90°N): polar stereographic grid, 2467x2367cells

This product is regional with a native grid based on conformal mapping and a very even spatial resolution. It is provided on a standard polar stereographic grid at 3 km resolution. All parameters are interpolated from the native Arakawa C grid model to the same output grid.



Vertical grid	50 hybrid layers
System Name	TOPAZ6
Model version	HYCOM v2.2.98 (ocean) CICE 5.1 (sea ice)
Horizontal output Resolution	3 km
Vertical output Resolution	Surface only
Interpolation method	Horizontally: bilinear.
Coupling	ESMF
Sea Ice Model	CICE v5.1 with 5 ice thickness categories
Bathymetry	GEBCO14
Initial conditions	WOA18 followed by 4 years of spinup.
Lateral boundary conditions	Global HR analysis and forecasting system. Tidal constituents from the FES2014 dataset.
Atmospheric and waves forcings	Source: ECMWF IFS HRES Frequency: 6-hourly Resolution: 1/10° Other: Completion of solar radiation daily cycle. Surface pressure from ECMWF (inverse barometer) is included.

	Wave terms (momentum, Stokes-Coriolis drift and vertical mixing) from the WAM 3km model ARCTIC_ANALYSISFORECAST_WAV_002_014
Tidal constituents	36 (as in FES2014)
Rivers	Climatology merged from Arctic-HYPE and E-HYPE reanalyses (SMHI, 2014). > 100 rivers Greenland mass loss from GRACE (GrIS CCI project). 35 terminal glaciers. Greenland surface mass balance from ECMWF reanalysis.
Data Assimilation scheme	None
Assimilated observations	N/A

Reference for the wave terms in HYCOM:

Ali, A., Christensen, K. H., Breivik, Ø., Malila, M., Raj, R. P., Bertino, L., Chassignet, E. and Bakhoday-Paskyabi, M. (2019). A comparison of Langmuir turbulence parameterizations and key wave effects in a numerical model of the North Atlantic and Arctic Oceans. *Ocean Modelling*, *137*, 76–97. https://doi.org/10.1016/j.ocemod.2019.02.005

c) Dataset details

Product Lines	ARCTIC_ANALYSISFORECAST_PHY_TIDE_002_015	
Geographical coverage	Arctic	
Datasets: dataset-topaz6-arc-15min-3km-be		
Product Type	Analysis	Forecast
Update frequency	Daily	Daily
Ensemble	None	None
Available time series	19/12/2017 to real-time	10-days forecast
Target delivery time	Following day at 01:00 UTC	Following day at 01:00 UTC
Temporal resolution (respective datasets)	15-minutes instantaneous	

Dataset : dataset-topaz6-arc-15min-3km-be Contains the 2D surface hourly variables			
Variable	**Unit**	**Name in file**	**Standard name**
Surface velocity in x-direction of polar stereographic grid Surface velocity in y-direction of polar stereographic grid	m/s m/s	vxo vyo	sea_water_x_velocity sea_water_y_velocity

Sea Surface Height	m	zos	sea_surface_height_above_geoid

d) Additional information on some parameters

zos [m]	sea_surface_height_above_geoid. The geoid is a surface of constant geopotential with which mean sea level would coincide if the ocean were at rest. The parameter "zos" is the difference between the actual sea surface height at any given time and place, and that which it would have if the ocean were at rest. The inverse barometer effect is included.
Vox, voy [m/s]	*The surface currents are formally "total surface currents" since the Stokes drift is included in June 2024, although the standard name is kept as "ocean currents" to avoid disrupting the aggregation server.*

e) Processing information

The model is integrated every day with updated atmospheric and waves forecasts

f) Catalogue updating

The catalogue is updated every day after midnight.

g) Time averaging

No temporal averaging is applied.

h) Product use and impact

The TOPAZ6 storm surge and tidal forecast (002_015) provides surface currents and sea surface heights in duplication with those in TOPAZ5 physical forecast (002_001). We expect these variables to be more accurate in TOPAZ6 near the coasts due to the higher resolution and the tidal and pressure processes but in the open ocean the data assimilation should give TOPAZ5 an advantage. We intend to close this topic by nudging TOPAZ6 to TOPAZ5 in November 2026.

3. FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4. FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

The ARC MFC products have been set up through different phases of Copernicus Marine and the file and dataset naming conventions have changed in the meantime. The table below shows correspondence between product name, dataset name, and file name.

Table 1: File name convention.

Product Name	*Dataset Name*	*File Name*
ARCTIC_ANALYSISFORECAST_PHY_TIDE_002_015	*dataset-topaz6-arc-15min-3km-be*	*20191214_qhr-metno-MODEL-topaz6-tide-ARC-b20191205-fv02.0.nc*

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

a. FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
dataset-topaz6-arc-15min-3km-be	*20191214_qhr-metno-MODEL-topaz6-tide-ARC-b20191205-fv02.0.nc*	2.79

b. STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5. REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-ARC-QUID-002-015.pdf

6. ANNEX

For dataset dataset-topaz6-arc-15min-3km-be

```
Sample file: netcdf \20200401_qhr-metno-MODEL-topaz6-tide-ARC-b20200401-fv02.0 {
dimensions:
        time = UNLIMITED ; // (96 currently)
        x = 2467 ;
        y = 2367 ;
variables:
        double time(time) ;
                time:standard_name = "time" ;
                time:long_name = "forecast time" ;
                time:units = "seconds since 1970-1-1T00:00:00Z" ;
                time:calendar = "standard" ;
                time:axis = "T" ;
        float longitude(y, x) ;
                longitude:standard_name = "longitude" ;
                longitude:long_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:_CoordinateAxisType = "Lon" ;
        float latitude(y, x) ;
                latitude:standard_name = "latitude" ;
                latitude:long_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:_CoordinateAxisType = "Lat" ;
        float x(x) ;
                x:standard_name = "projection_x_coordinate" ;
                x:units = "100  km" ;
                x:axis = "X" ;
        float y(y) ;
                y:standard_name = "projection_y_coordinate" ;
                y:units = "100  km" ;
                y:axis = "Y" ;
        int stereographic ;
                stereographic:grid_mapping_name = "polar_stereographic" ;
                stereographic:latitude_of_projection_origin = 90. ;
                stereographic:longitude_of_projection_origin = -45. ;
                stereographic:scale_factor_at_projection_origin = 1. ;
                stereographic:straight_vertical_longitude_from_pole = -45. ;
                stereographic:earth_redius = 6378273. ;
                stereographic:proj4 = "+proj=stere +lon_0=-45 +lat_0=90 +k=1 +R=6378273
+no_defs" ;
                stereographic:false_easting = 0. ;
                stereographic:false_northing = 0. ;
        float model_depth(y, x) ;
                model_depth:standard_name = "sea_floor_depth_below_sea_level" ;
                model_depth:units = "meter" ;
                model_depth:grid_mapping = "stereographic" ;
                model_depth:coordinates = "latitude longitude" ;
                model_depth:_FillValue = -1.e+14f ;
```

```
                model_depth:missing_value = -1.e+14f ;
                model_depth:cell_methods = "area: mean" ;
        float vxo(time, y, x) ;
                vxo:standard_name = "sea_water_x_velocity" ;
                vxo:units = "m s-1" ;
                vxo:grid_mapping = "stereographic" ;
                vxo:coordinates = "latitude longitude" ;
                vxo:_FillValue = -1.e+14f ;
                vxo:missing_value = -1.e+14f ;
                vxo:cell_methods = "area: mean" ;
        float vyo(time, y, x) ;
                vyo:standard_name = "sea_water_y_velocity" ;
                vyo:units = "m s-1" ;
                vyo:grid_mapping = "stereographic" ;
                vyo:coordinates = "latitude longitude" ;
                vyo:_FillValue = -1.e+14f ;
                vyo:missing_value = -1.e+14f ;
                vyo:cell_methods = "area: mean" ;
        float zos(time, y, x) ;
                zos:standard_name = "sea_surface_height_above_geoid" ;
                zos:units = "m" ;
                zos:grid_mapping = "stereographic" ;
                zos:coordinates = "latitude longitude" ;
                zos:_FillValue = -1.e+14f ;
                zos:missing_value = -1.e+14f ;
                zos:cell_methods = "area: mean" ;

// global attributes:
                :CDI = "Climate Data Interface version 1.9.2 (http://mpimet.mpg.de/cdi)"
;
                :history = "20200401:Created by program hyc2proj, version V0.3" ;
                :source = "NERSC-HYCOM model fields" ;
                :institution = "NERSC, Thormoehlens gate 47, N-5006 Bergen, Norway" ;
                :Conventions = "CF-1.4" ;
                :references = "http://topaz.nersc.no" ;
                :field_type = "Files based on file type archs" ;
                :field_date = "2020-04-01T00:00:00Z" ;
                :CDO = "Climate Data Operators version 1.9.2
(http://mpimet.mpg.de/cdo)" ;
                :bulletin_date = "2020-04-01" ;
                :title = "Arctic Ocean Physics Analysis and Forecast, 3 km quarter-hourly
instantaneous (dataset-topaz6-tide-arc-qhr-myoceanv2-be)" ;
                :bulletin_type = "Forecast" ;
                :Forecast_range = "10 days" ;
}
```